

■ 1.1.2 Exact and Approximate Results

A standard electronic calculator does all your calculations to a particular accuracy, say ten decimal digits. With *Mathematica*, however, you can often get *exact* results.

Mathematica gives an *exact* result for 2^{100} ,
even though it has 31 decimal digits.

```
In[1]:= 2 ^ 100
Out[1]= 1267650600228229401496703205376
```

You can tell *Mathematica* to give you an approximate numerical result, just like a calculator, by ending your input with `//N`.

The `N` stands for “numerical”. It must be a capital letter. Section 2.1 will explain what the `//` means.

This gives an approximate numerical result.

```
In[2]:= 2 ^ 100 //N
Out[2]= 1.26765 1030
```

Mathematica can give results in terms of rational numbers.

```
In[3]:= 1/3 + 2/7
Out[3]=  $\frac{13}{21}$ 
```

`//N` always gives the approximate numerical result.

```
In[4]:= 1/3 + 2/7 //N
Out[4]= 0.619048
```

`expr //N` give an approximate numerical value for `expr`

Getting numerical approximations.

When you type in an integer like 7, *Mathematica* assumes that it is exact. If you type in a number like 4.5, with an explicit decimal point, *Mathematica* assumes that it is accurate only to a fixed number of decimal places.

This is taken to be an exact rational number, and reduced to its lowest terms.

```
In[5]:= 452/62
Out[5]=  $\frac{226}{31}$ 
```

Whenever you give a number with an explicit decimal point, *Mathematica* produces an approximate numerical result.

```
In[6]:= 452.3/62  
Out[6]= 7.29516
```

Here again, the presence of the decimal point makes *Mathematica* give you an approximate numerical result.

```
In[7]:= 452./62  
Out[7]= 7.29032
```

When any number in an arithmetic expression is given with an explicit decimal point, you get an approximate numerical result for the whole expression.

```
In[8]:= 1. + 452/62  
Out[8]= 8.29032
```